Do the following exercises from Chapter 6 of the textbook: 10, 14, and 21. Can scan in information as appropriate and add to MS Word document for submission.

1.

10.

K. Nelson, project manager of Print Software, Inc., wants you to prepare a project network; compute the early, late, and slack activity times; determine the planned project duration; and identify the critical path. His assistant has collected the following information for the color printer drivers software project: page 199

ID	Description	Predecessor	Time
A	External specifications	None	8
B	Review design features	A	2
C	Document new features	A	3
D	Write software	A	60
E	Program and test	B	40
F	Edit and publish notes	C	2
G	Review manual	D	2
H	Alpha site	E, F	20
I	Print manual	G	10
J	Beta site	H, I	10
K	Manufacture	J	12
L	Release and ship	K	3

Ans

ID	Duration	Predecessors
A	8	None
B	2	A
C	3	A
D	60	A
E	40	B
F	2	C
G	2	D
H	20	E, F
I	10	G
J	10	H, I
K	12	J
L	3	K

$$EF\ (Early\ Finish)\ =\ ES + Duration$$

$$LS\ (Late\ Start)\ =\ LF - Duration$$

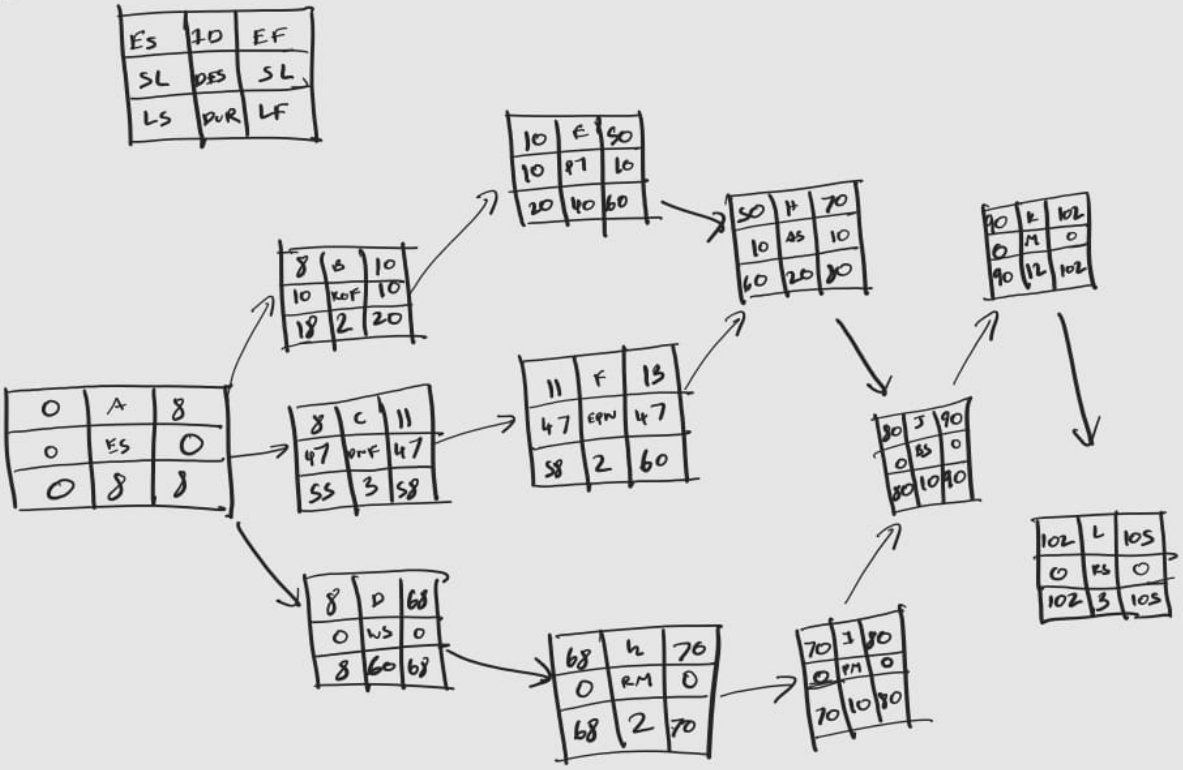
$$Slack = LS - ES$$

Summary Table

ID	ES	EF	LS	LF	Slack	Critical Path?
A	0	8	0	8	0	Yes
B	8	10	18	20	10	No
C	8	11	55	58	47	No
D	8	68	8	68	0	Yes
E	10	50	20	60	10	No
F	11	13	58	60	47	No
G	68	70	68	70	0	Yes
H	50	70	60	80	10	No
I	70	80	70	80	0	Yes
J	80	90	80	90	0	Yes
K	90	102	90	102	0	Yes
L	102	105	102	105	0	Yes

- Activities with 0 slack are on the critical path.
- The critical path: A → D → G → I → J → K → L
- Project duration: 105 days

10.



Burst → A

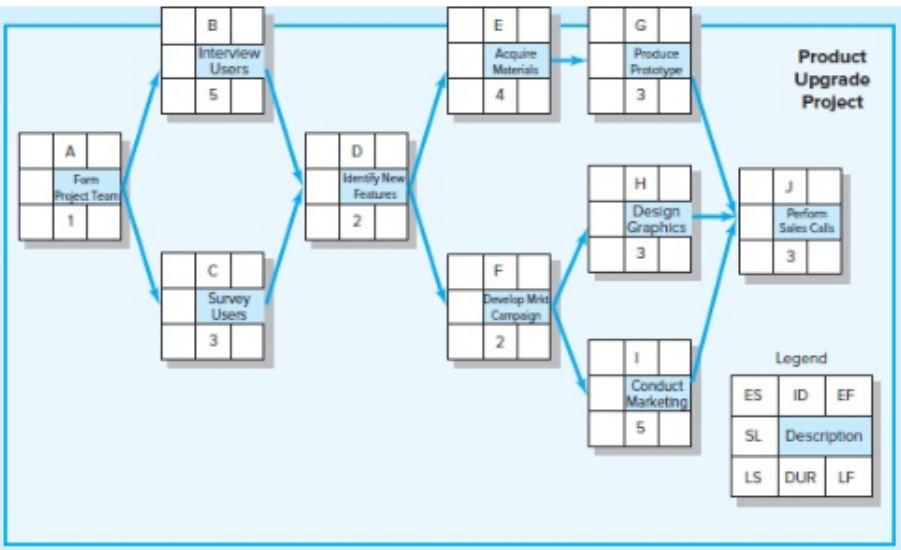
Merge → H, J

Critical path → A → D → G → I → J → K → L

2.

14.

You are managing a product upgrade project for Bangkokagogo. Given the project network that follows, complete the forward and backward pass, compute activity slack, and identify the critical path. Use this information to create a Gantt chart for the project. Be sure to show slack for noncritical activities.



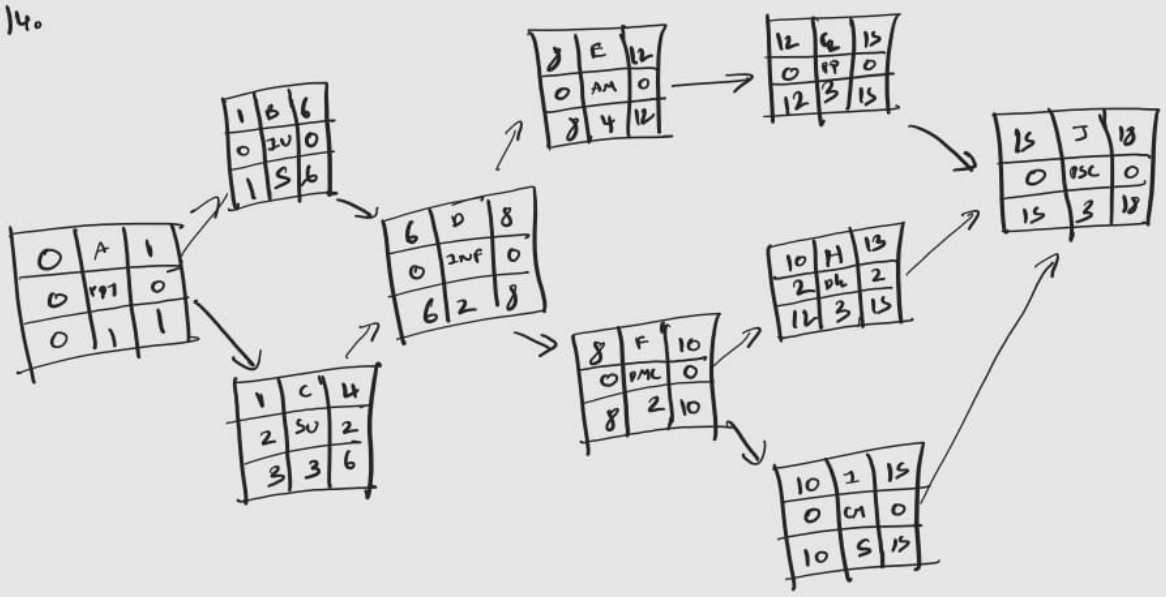
page 201

Product Upgrade Project Gantt Chart

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Form Project Team																		
Interview Users																		
Survey Users																		
ID New Features																		
Acquire Materials																		
Dev Mrkt Campaign																		
Produce Prototypes																		
Design Graphics																		
Conduct Marketing																		
Perform Sales Calls																		

Ans

14.



Burst → A

Merge → D, J

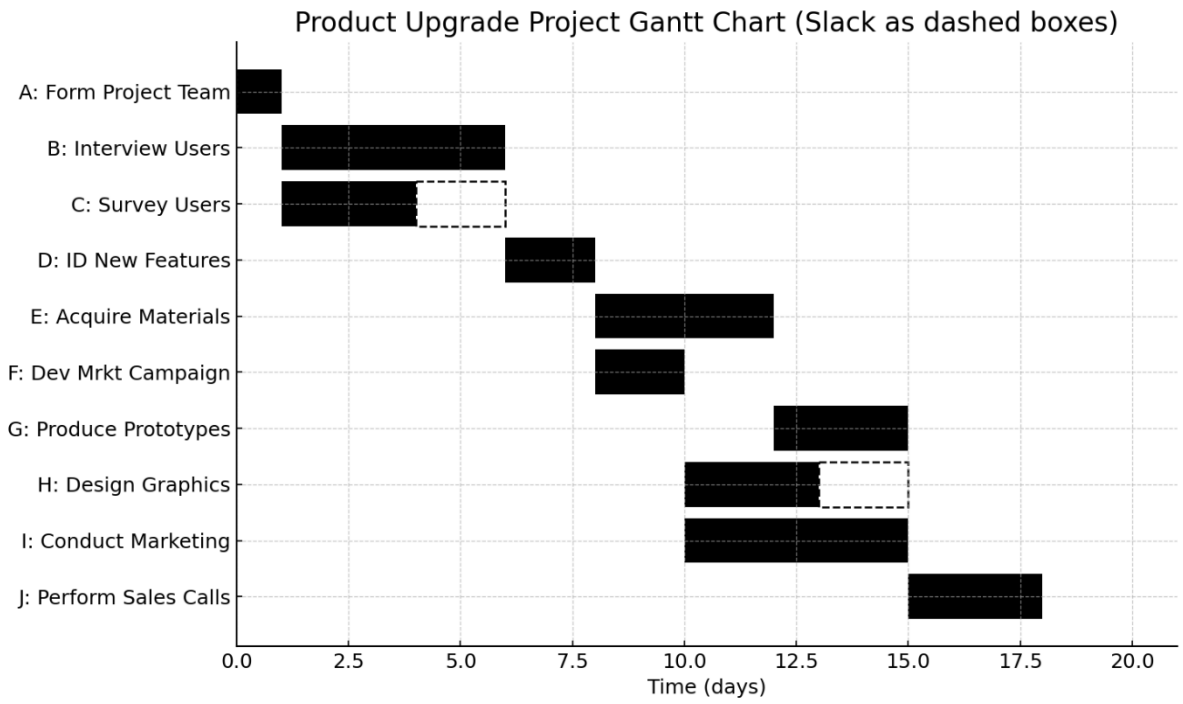
Critical path → A → B → D → E → H → J

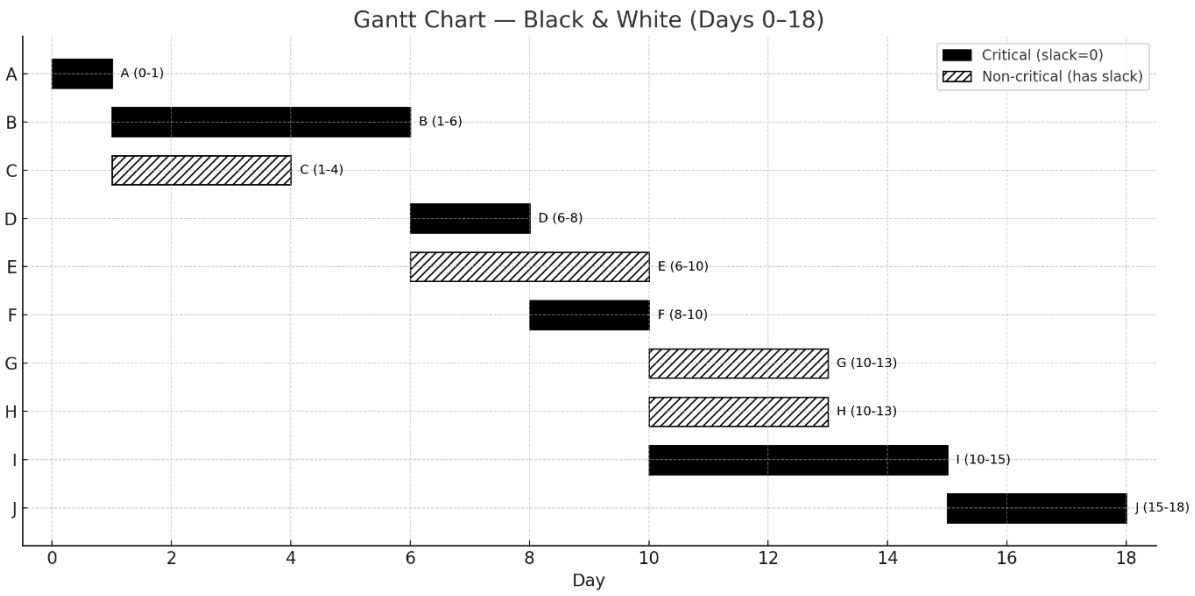
(or)

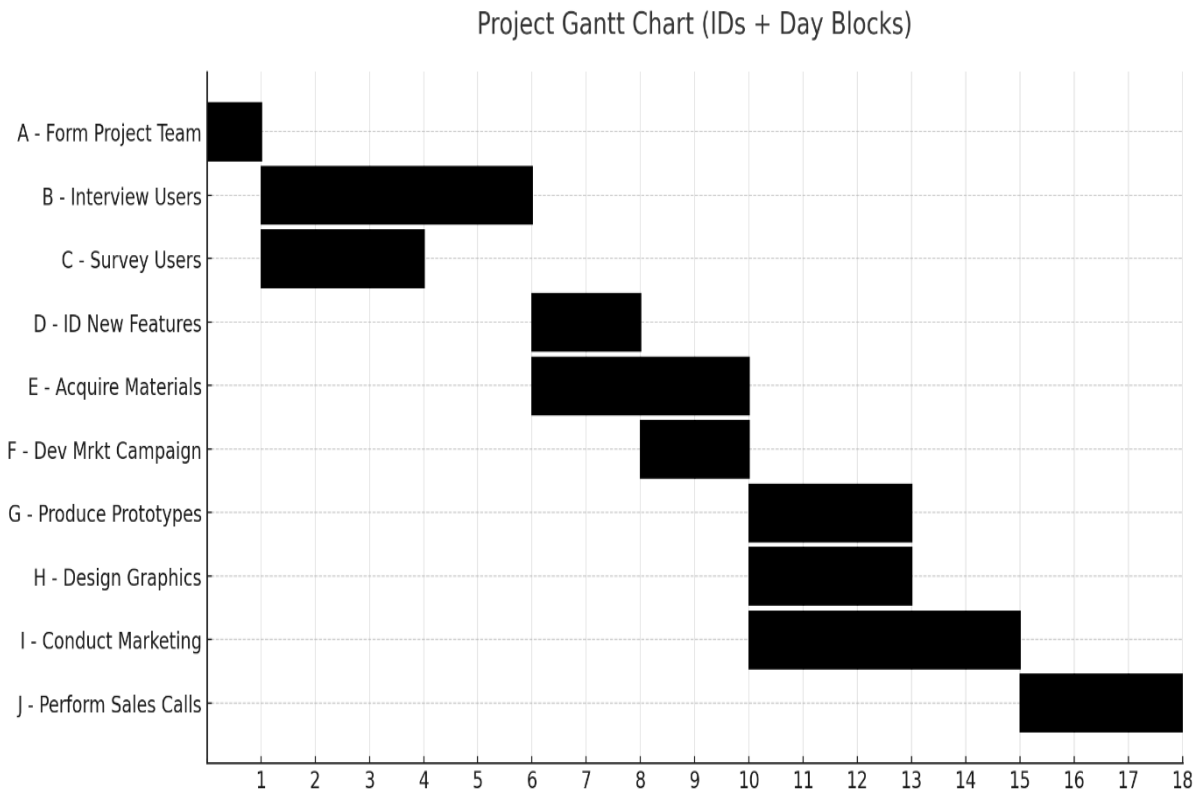
A → B → D → F → I → J

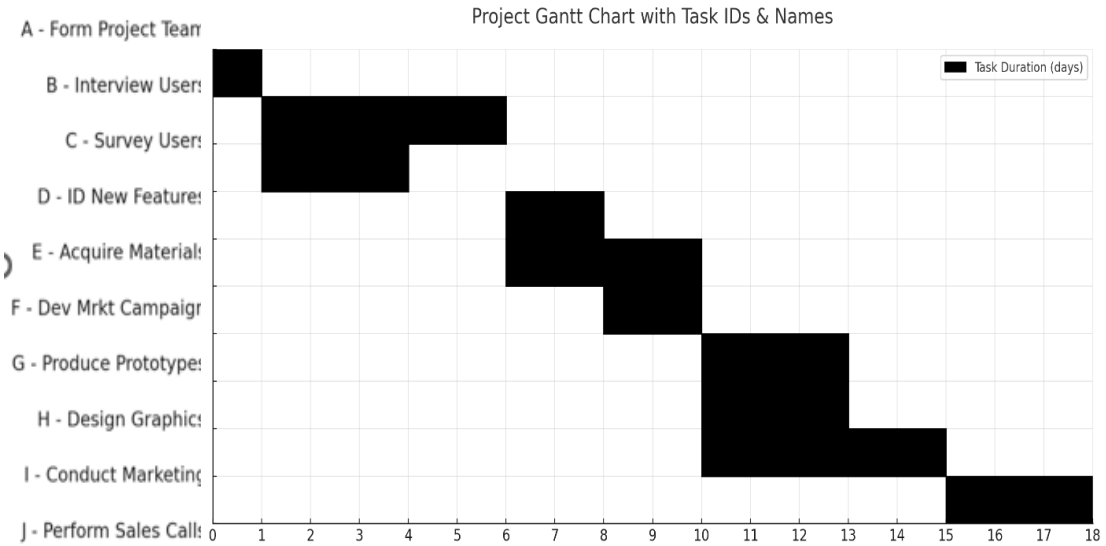
Slack Activities → C → 2 units

H → 2 units


Product Upgrade Project Gantt Chart (Slack as dashed boxes)
A: Form Project Team
B: Interview Users
C: Survey Users
D: ID New Features
E: Acquire Materials
F: Dev Mrkt Campaign
G: Produce Prototypes
H: Design Graphics
I: Conduct Marketing
J: Perform Sales Calls
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time (days)


Gantt Chart — Black & White (Days 0–18)
Critical (slack=0)
Non-critical (has slack)
A
B
C
D
E
F
G
H
I
J
A (0-1)
B (1-6)
C (1-4)
D (6-8)
E (6-10)
F (8-10)
G (10-13)
H (10-13)
I (10-15)
J (15-18)
0
2
4
6
8
10
12
14
16
18
Day

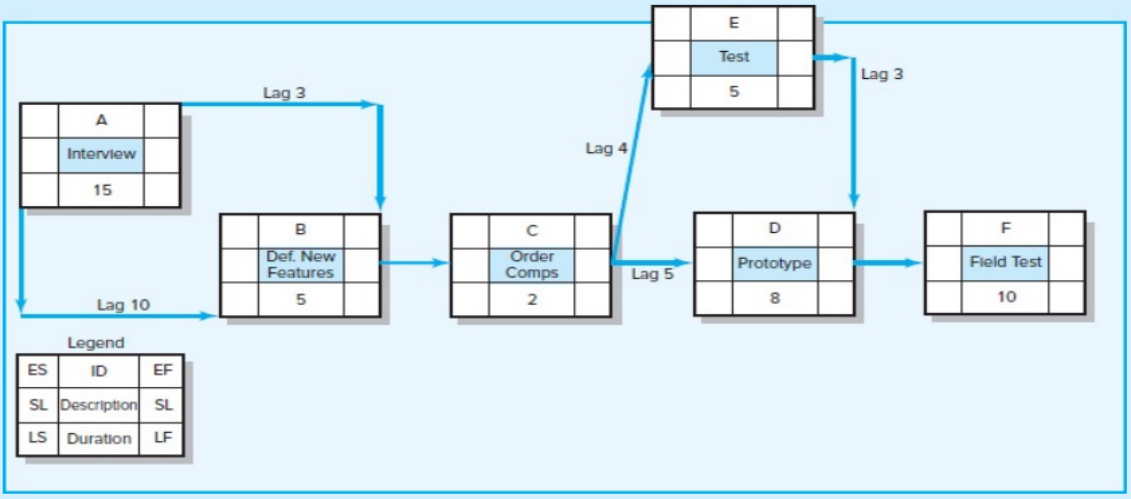

Project Gantt Chart (IDs + Day Blocks)
A - Form Project Team
B - Interview Users
C - Survey Users
D - ID New Features
E - Acquire Materials
F - Dev Mrkt Campaign
G - Produce Prototypes
H - Design Graphics
I - Conduct Marketing
J - Perform Sales Calls
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18


Project Gantt Chart with Task IDs & Names
Task Duration (days)
A - Form Project Tean
B - Interview Users
C - Survey Users
D - ID New Features
E - Acquire Materials
F - Dev Mrkt Campaign
G - Produce Prototypes
H - Design Graphics
I - Conduct Marketing
J - Perform Sales Calls
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18

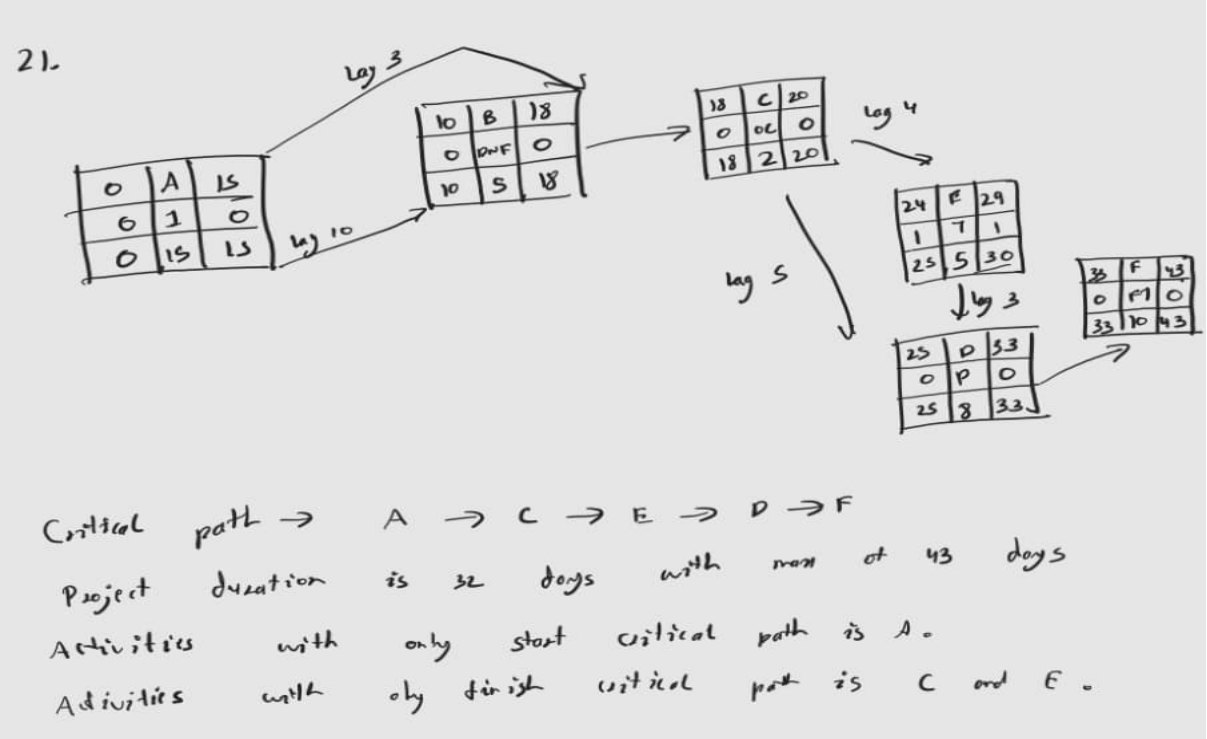
3.

21.

21. Given the information in the following lag exercises, compute the early, late, and slack times for the project network.* Which activities on the critical path have only the start or finish of the activity on the critical path? page 205

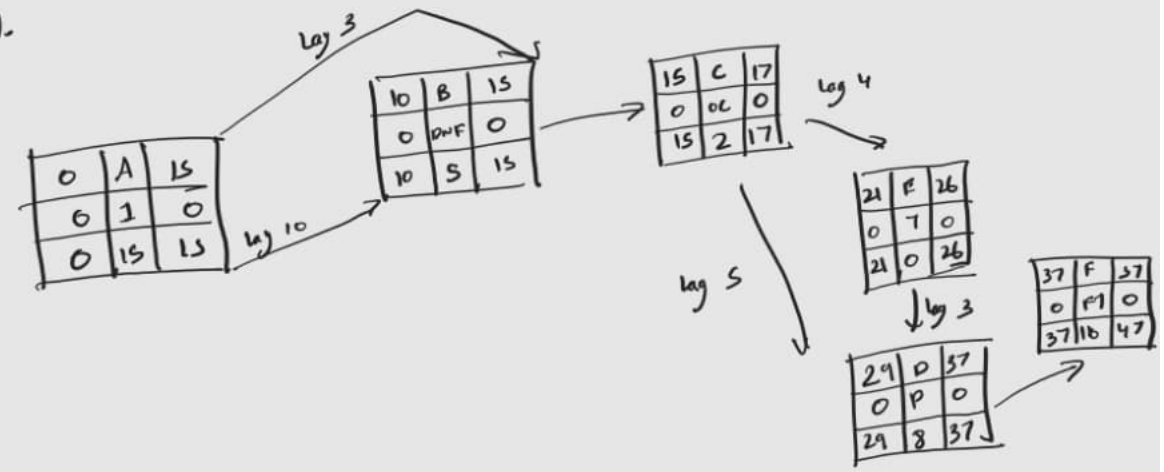


Ans



OR

21.



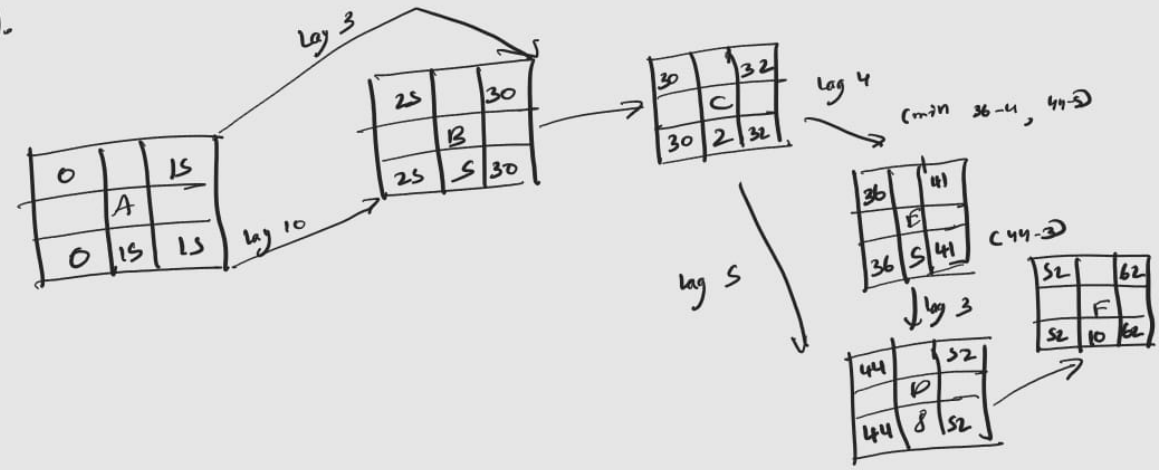
Critical path → A → B → C → E → D → F

A → B have only the start and finish of the activity on critical path

BC, ED are fully critical

OR

21.



Critical path → A → B → C → E → D → F

Both the start and the finish of each activity lie on the critical path. An activity would have only its start (or only its finish) on the critical path only if ES = LS but EF ≠ LF (or EF = LF but ES ≠ LS), which would give nonzero slack — not the case here.